GREEN VALLEY EDUCATIONAL INSTITUTE (GVEI)

DEPARTMENT OF INFORMATION & COMMUNICATION TECHNOLOGY (ICT)

AI, CODING, INNOVATION & ENRICHMENT ACTIVITIES (2026–2027)

Prepared by: Mr. Mehraj U Din Mir, HOD ICT **Designation:** Head of Department – ICT

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1. Strategic Vision: From Digital Literacy to Future-Ready Competence

The ICT Department at Green Valley Educational Institute (GVEI) is committed to shifting students from passive consumers of technology to informed, creative, responsible, and capable computational innovators. Our pedagogical framework empowers learners to design practical solutions, decompose complex problems, and apply technological skills meaningfully.

Artificial Intelligence, programming, robotics, and innovation initiatives directly enrich classroom learning by providing continuous opportunities for applied experimentation, algorithmic thinking, interdisciplinary collaboration, and independent creation.

> **THE THREE-PILLAR STRATEGIC MANDATE**
> *The departmental enrichment paradigm targets three interconnected core outcomes:*
> • *Understand: Cultivate deep digital, computational, and AI literacy calibrated to students' developmental stages.*
> • *Create: Apply technical knowledge through code authoring, multimedia artifacts, engineering prototypes, and algorithmic problem-solving.*
> • *Innovate Responsibly: Critically critique technology, deploy AI ethically, and design solutions with profound consideration for societal well-being, accessibility, and environmental sustainability.*

This policy document establishes a structured framework for documenting departmental enrichment drives, student participation registers, mentor allocations, project portfolios, verified competition achievements, and demonstrable learning outcomes across all school divisions.

2. Objectives of AI, Coding and Innovation Enrichment

Departmental enrichment initiatives are designed to achieve eight foundational pedagogical objectives:

2.1 Age-Appropriate Emerging Tech Mastery: Develop progressive understanding of Artificial Intelligence, cloud systems, and modern digital architectures.

2.2 Computational Thinking Rigor: Cultivate logical reasoning, pattern recognition, algorithmic decomposition, iterative abstraction, and systematic debugging.

2.3 Applied Design & Rapid Prototyping: Provide regular opportunities to design, build, test, and optimize software applications and physical computing prototypes.

2.4 Inquiry & Problem-Solving Habits: Nurture creativity, intellectual curiosity, and autonomous problem-solving through authentic challenge briefs.

2.5 Ethical Tech Stewardship & AI Literacy: Instill digital citizenship, data privacy awareness, academic integrity, and rigorous verification of AI-generated content.

2.6 Interdisciplinary Synthesis: Connect computational concepts with environmental science, physical sciences, mathematics, and community problem-solving.

2.7 Public Demonstration & Competitions: Furnish platforms for students to showcase projects via exhibitions, hackathons, coding symposiums, and peer presentations.

2.8 Talent Identification & Mentorship: Identify nascent student aptitudes early and provide structured mentorship to foster continuous technical growth.

Operational Directive: All enrichment activities must remain strictly calibrated to approved curriculum goals, cognitive readiness, laboratory hardware capacity, and the overarching academic priorities of GVEI.

3. AI Literacy and Responsible AI

AI literacy enables learners to comprehend the capabilities and algorithmic mechanics of modern AI systems, identify their technical and ethical limitations, and deploy relevant tools with discernment. Instruction is developmentally staged and does not presume universal access to generative models.

Focus Area	Illustrative Learning Activities	Verifiable Evidence of Learning
Understanding AI Mechanics	Demystifying rule-based systems vs. machine learning; identifying real-world AI applications (search, navigation, recommendation).	Concept mapping worksheets, system comparison logs, and student oral explanations.
Data, Features & Patterns	Interactive classification tasks; understanding training datasets, labeling, and feature extraction using visual tools.	Teachable Machine training logs, classification charts, and reflective student journals.
Human Agency & Decision-Making	Case studies analyzing automated decision-making errors, hallucination in LLMs, and situations requiring human oversight.	Case analysis briefs, structured Socratic debate notes, and critical essay summaries.
Responsible & Ethical AI	Analyzing algorithmic bias, copyright boundaries, deepfakes, misinformation, and personal data harvesting.	Scenario evaluation matrices, student-authored ethics codes, and cyber safety posters.
Generative AI Literacy	Structured prompt formulation, cross-referencing AI outputs against trusted sources, and fact-checking generated text/code.	Documented prompt logs, side-by-side fact-checking dossiers, and reflection sheets.
AI for Social Good	Designing conceptual AI solutions addressing environmental conservation, accessibility, local agriculture, or disaster response.	Project design briefs, user journey wireframes, and prototype slide decks.

> **AI ETHICAL SAFEGUARDS**
> *Strict Governance Safeguard: AI activities must reinforce factual verification, human agency, student data privacy, and academic integrity. Under no circumstances may student personal records or identifiable information be submitted into public generative AI engines.*

4. Coding, Robotics and Computational Thinking

Coding and robotics activities cultivate the capacity to translate abstract thoughts into structured procedural algorithms, test solutions through iterative execution, and refine software architectures using objective feedback.

Enrichment Strand	Core Pedagogical Learning Focus	Mandatory Evidentiary Output
Block-Based Coding (Primary/Middle)	Algorithmic sequencing, event triggers, nested loops, conditional logic, and multi-sprite	Scratch project repositories (.sb3), animation recordings, and logic task

	interactions.	cards.
Text-Based Programming (Senior)	Python syntax, variable typing, conditional branching, iteration, modular functions, and robust exception handling.	Python source files (.py), trace-table notes, terminal bug logs, and execution records.
Computational Thinking (CT)	Problem decomposition, pattern recognition, abstraction, flowcharting, and pseudocode formulation.	Algorithmic flowchart sheets, trace tables, unplugged logic puzzle portfolios.
Robotics & Physical Computing	Input-process-output cycles, motor control, ultrasonic/infrared sensor interfacing, and physical automated loops.	Robot run logs, hardware wiring diagrams, demonstration videos, and build logs.
Digital Product Engineering	End-to-end design thinking: user requirement scoping, UI wireframing, sprint testing, and usability refinement.	Product specification sheets, git commit logs, user feedback rubrics, and final builds.
Collaborative Problem-Solving	Agile task division, peer pair-programming, code review circles, and shared software project management.	Team sprint logs, task delegation boards, individual contribution statements, and peer reviews.

Pedagogical Standard: Departmental evaluation prioritizes deep conceptual logic and the student's ability to verbally explain and debug their code—not merely the superficial appearance of the output.

5. AI & Innovation Club: Institutional Enrichment Framework

The AI & Innovation Club functions as an institutional platform where students apply advanced computing concepts, collaborate on multi-week projects, and prepare for state and national hackathons.

Club efficacy is proven through tangible student deliverables, consistent attendance, structured mentor guidance, and verified competition results—never through nominal club rosters.

5.1 Core Club Mandates

- Practical Skill Acceleration: Deepening hands-on proficiencies in Python programming, robotics, UI design, and machine learning models.
- Real-World Problem Scoping: Empowering students to identify authentic school or community challenges and engineer workable digital solutions.
- Interdisciplinary Synthesis: Integrating computer science with environmental conservation, mathematics, and assistive technologies.
- External Hackathon Preparation: Systematically preparing teams for inter-school tech symposiums, ATL marathons, and coding Olympiads.
- Project Management Mastery: Instilling agile development habits, milestone tracking, division of labor, and code documentation.
- Individualized Mentorship: Pairing students with faculty mentors based on specialized interest domains and technical readiness.
- Ethical Innovation: Reinforcing data ethics, intellectual property respect, and socially responsible technology stewardship.

5.2 Annual Activity Calendar (2026–2027)

Academic Period	Strategic Club Focus	Proposed Operational Activities	Expected Tangible Deliverable
April – June 2026	**Planning, Induction & Skill Audits**	Formulate club charters, conduct student orientation, audit laboratory hardware, and administer skill-interest surveys.	Approved club roadmap, mentor allocation charts, and baseline student skill profiles.

July – September 2026	**Skill Building & Hackathon Sprints**	Conduct hands-on coding clinics, introductory AI literacy sessions, robotics sensor labs, and team project ideation.	Active participation registers, project specification briefs, and preliminary code prototypes.
October – December 2026	**Project Maturation & Testing**	Execute multi-week sprint cycles, conduct peer code reviews, debug hardware prototypes, and hold mid-session reviews.	Functional software builds, hardware wireframes, user testing logs, and sprint milestone sign-offs.
January – March 2027	**Exhibition, Defense & Review**	Host the Annual GVEI Tech Innovation Expo, submit entries to regional hackathons, and compile project portfolios.	Final project showcases, competition award dossiers, annual audit reports, and next-session roadmaps.

6. Student Enrolment, Skill Mapping and Mentor Allocation

The club maintains a live tracking register mapping student profiles, interest tracks, developmental goals, and assigned faculty mentors:

Student Name / ID	Class / Wing	Specialized Interest	Prior Technical Exposure	Target Development Goal	Assigned Faculty Mentor
Ayan Ahmad (GVEI-842)	Class XI-CS	Machine Learning & AI	Python syntax & basic OOP	Train neural vision model for leaf disease	**Mr. Mehraj U Din Mir**
Zainab Bilal (GVEI-905)	Class X-A	Web Apps & Database	HTML/CSS & basic SQL	Build student book exchange portal	**Faculty Member A**
Mohammad Umar (GVEI-711)	Class IX-B	Robotics & Microcontrollers	Block coding & basic electronics	Program obstacle-avoiding smart cart	**Faculty Member B**
Bareen Zahra (GVEI-654)	Class VIII-C	Game Development	Scratch animations & loops	Develop multi-level math quiz game	**Faculty Member C**
Faizan Tariq (GVEI-512)	Class VII-A	UI/UX & Graphic Design	Canva & presentation design	Create digital cyber safety comic strip	**Faculty Member D**
Syed Musaib (GVEI-922)	Class XII-CS	Data Science & Pandas	Python lists, tuples, SQL	Develop automated student attendance analytics	**Mr. Mehraj U Din Mir**

6.2 Mentor Allocation Governance

Mentors are assigned based on specialized expertise and student readiness. The mentorship framework enforces:

- **Dedicated Mentor Pairing:** Each project team is paired with an accountable mentor overseeing weekly progress.
- **Milestone Checkpoints:** Structured bi-weekly sprint reviews assessing wireframes, code commits, and testing logs.
- **Individual Contribution Audits:** Ensuring that within collaborative teams, every student's code and logic contribution remains transparent and identifiable.

6.3 Attendance & Participation Tracking

The club maintains a strict attendance and engagement log. Mere attendance does not prove competence; active contribution to project repositories is mandatory for credentialing.

7. Projects, Competitions, Workshops and Demonstrations

7.1 Project-Based Learning Lifecycle

Student engineering projects follow a disciplined 7-stage development lifecycle:

7.1.1 Problem Scoping & Opportunity Scoping: Identifying an authentic social, school, or community challenge.

7.1.2 Specification & Metrics Definition: Defining functional requirements, edge cases, and objective success metrics.

7.1.3 Algorithmic Research & Architecture: Decomposing problems into flowcharts, data schemas, and modular functions.

7.1.4 Iterative Prototyping: Authoring working code or constructing physical hardware circuits.

7.1.5 Rigorous Testing & Error Diagnosis: Executing edge-case testing, diagnosing compiler bugs, and refining logic.

7.1.6 Technical Defense & Showcase: Presenting functional deliverables and defending architectural choices.

7.1.7 Critical Reflection & Future Scope: Documenting bottlenecks overcome, technical limitations, and future roadmap.

7.2 External Competitions & Hackathons

The department actively coordinates student participation in inter-school hackathons, CBSE science symposiums, and national coding competitions. Archival records strictly distinguish between Participation, Selection, Finalist Qualification, and Award-Winning Placements.

7.3 Technical Workshops & Demonstrations

Enrichment workshops conducted by faculty or invited experts are fully documented, recording session objectives, software environments, student participation logs, and resulting digital artifacts.

8. Project Evaluation and Innovation Quality

Projects are assessed using a standardized, 8-criterion rubric emphasizing algorithmic reasoning, problem-solving, and responsible tech stewardship over cosmetic design:

Evaluation Criterion	Assessment Focus & Student Capability Evaluated	Weight
Problem Understanding	Clear identification, scoping, and justification of the targeted challenge.	**10%**
Conceptual Rigor	Accurate, robust application of computing concepts, data structures, and syntax.	**20%**
Operational Functionality	Execution reliability: whether the software/hardware runs without crashing.	**20%**
Computational Thinking	Evidence of systematic decomposition, efficient algorithms, and structured debugging.	**15%**
Creativity & Originality	Original student ideation, custom logic formulation, and novel user workflows.	**10%**
Testing & Iterative Refinement	Documented bug logs, sprint commit history, and improvements post-feedback.	**10%**
Responsible Innovation	Proactive protection of privacy, accessibility compliance, and ethical data handling.	**5%**
Technical Defense & Defense	Student capability to verbally articulate code structure, explain functions, and defend choices.	**10%**

Rubric Scale: Beginning (1) | Developing (2) | Proficient (3) | Advanced (4). Rubrics are shared with students at project inception.

9. Student Achievements and Verified Outcomes

The department maintains a verified registry of student competition milestones, hackathon achievements, and capstone software deployments:

Student / Team	Class	Activity / Event Title	Verifiable Achievement / Outcome	Primary Verification Evidence	Date
CodeDay Srinagar Team	Class XI/XII	CodeDay Hackathon (NIT Srinagar)	**Best Innovative Project Award**	Official Certificate & Project Repo	2026-06-28
Senior Robotics Squad	Class X	Inter-School Tech Expo 2026	**First Runner-up (Automated Systems)**	Trophy, Citation & Event Notice	2026-08-14
Ayan Ahmad & Team	Class XI	Kashmir AI Innovation Challenge	**Finalist Qualification (Top 5 Teams)**	Selection Letter & Build Dossier	2026-09-18
Middle Wing Coding Club	Class VII/VIII	National Scratch Challenge	**Exemplary Creative Coding Badge**	Official Digital Merit Badges	2026-07-10
Syed Musaib	Class XII	CBSE Regional Student Project	**Selected for State Level Showcase**	CBSE Official Verification Notice	2026-09-25
Junior Digital Artists	Class IV/V	Digital Art & Cyber Awareness Week	**Institutional Gold & Silver Medals**	Departmental Certificates & Files	2026-05-20

Veracity Protocol: All achievement claims must be backed by official certificates, award gazettes, or verified project repositories. Participation or finalist status must never be conflated with an outright victory.

10. Evidence Repository and Documentation Standards

The department maintains a centrally organized evidence archive documenting student learning, development iterations, and mentor reviews:

- **Session Dossiers:** Detailed session plans, educator attendance logs, and student participation rosters.
- **Source Code Repositories:** Archived Python scripts, SQL database dumps, and Scratch project source files.
- **Engineering Portfolios:** Project specification briefs, milestone logs, bug tracing sheets, and completed rubrics.
- **Photographic Evidence:** High-resolution photos documenting live hacking sessions, robotics testing, and expo displays.
- **Award Documentation:** Digitally archived certificates of merit, jury score sheets, and official letters.

11. Outcomes and Demonstrated Competencies

Enrichment programmes are evaluated through observable competency benchmarks rather than superficial attendance figures:

Targeted Competency Outcome	Observable Evidence Audited in Review
AI & Digital Literacy	Student oral defenses, scenario case-study analyses, and documented prompt engineering logs.
Computational Thinking (CT)	Structured pseudocode, detailed flowcharts, and demonstrable code tracing competence.
Practical Software Engineering	Execution of fully functional, bug-free software applications, scripts, and databases.
Creative Prototyping	Multi-iteration UI wireframes, graphic designs, and custom algorithmic game mechanics.
Collaborative Engineering	Documented task allocation, peer pair-programming logs, and shared project builds.
Responsible Technology Use	Strict compliance with data privacy, source attribution, and ethical AI usage guidelines.

Autonomous Problem-Solving	Student-initiated debugging, refactoring, and independent resolution of compiler errors.
Real-World Impact Scoping	Functional software solutions addressing school operations, sustainability, or local community needs.

12. Monitoring, Evaluation and Continuous Improvement

The Head of Department and designated club mentors review enrichment outcomes monthly to identify instructional hurdles, refine mentoring, and allocate hardware resources:

Enrichment Track	Targeted Strategic Outcome	Current Empirical Status	Available Evidence	Identified Bottleneck	Corrective Action Planned
Python Coding Clinic	Mastery of modular functions & OOP	**Completed Phase 1**	Source repos & task sheets	Syntax errors in nested logic	Zero-period code tracing drills
AI Literacy Workshops	Ethical AI understanding & prompt literacy	**In Progress**	Prompt logs & essays	Over-reliance on LLM code	Strict human-verification mandate
Robotics Prototyping	Functional automated sensor circuits	**Hardware Tested**	Circuit wireframes & video	Limited sensor component kits	Procurement of 5 Arduino kits
Game Design (Scratch)	Interactive multi-level game loops	**Completed Phase 2**	Project .sb3 files & rubrics	Collision event timing bugs	Peer code-review workshop
Hackathon Training	Regional tech competition readiness	**Active Mentoring**	Sprint logs & build repos	Time management in sprints	Conducted 3-hour timed hackathon

13. Evidence File Index for the Academic Audit

The department maintains nine primary documentary files to support academic inspection:

13.1 Strategic Enrichment Roadmap: Annual charter, pedagogical vision, resource plans, and approved calendar.

13.2 AI Literacy & Ethics Dossier: Curriculum modules, prompt verification sheets, and student ethics essays.

13.3 Coding, Robotics & CT Repository: Task cards, student source code, terminal trace logs, and assessment records.

13.4 AI & Innovation Club Governance: Membership rosters, skill-interest profiles, mentor pairings, and attendance logs.

13.5 Capstone Projects & Innovation: Project briefs, iterative sprint logs, working prototypes, and evaluation rubrics.

13.6 Workshops & Hackathons Archive: Faculty workshop reports, competition submissions, and external event rosters.

13.7 Student Achievements & Certifications: Official certificates of merit, medal citations, and verified competition results.

13.8 Demonstrated Learning Outcomes: Longitudinal skill tracking charts, peer reviews, and student reflection journals.

13.9 Continuous Improvement Register: Monthly review minutes, barrier analyses, and corrective pedagogical interventions.

14. Priority Evidence Sets for the Academic Audit

For academic audit inspections, the department maintains three benchmark evidence dossiers readily accessible:

ACADEMIC AUDIT PRIORITY EVIDENCE SETS

- *Evidence Set 1 — AI & Innovation Club in Practice: Approved charter, active student membership roster, mentor pairing register, sample activity brief, and verified student code outputs.*
- *Evidence Set 2 — Student Project from Ideation to Output: Comprehensive specification brief, algorithmic flowchart, working software build/prototype, completed evaluation rubric, and student defense reflection.*
- *Evidence Set 3 — Verified Achievement & Skill Development: Official competition award certificate or longitudinal student progress case study proving baseline deficiency, intervention, and post-support mastery.*

15. Expected Outcome and Departmental Commitment

The ICT Department establishes AI, programming, robotics, and innovation enrichment as a purposeful, rigorous extension of classroom instruction. It transforms students from conceptual observers into creative computational practitioners capable of authoring functional code, engineering robust prototypes, and critically evaluating emerging technologies.

Departmental Commitment: The ICT Department commits to promoting authentic, responsible student innovation, maintaining verifiable empirical records of engagement and achievement, and deploying evidence from enrichment drives to continually elevate teaching practices and student learning opportunities at Green Valley Educational Institute.

PREPARED AND SUBMITTED BY:
Mr. Mehraj U Din Mir, HOD ICT
Head of Department – ICT
Green Valley Educational Institute (GVEI), Srinagar

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